

Duality

Lecture 6

October 8, 2026

Quiz

Consider the following primal **maximization** problem:

$$\begin{array}{ll} (P) & \max \quad x_1 + x_2 \\ & \text{s.t.} \quad x_1 - x_2 \leq -1, \\ & \quad \quad -x_1 + x_2 \leq -1, \\ & \quad \quad x_1, x_2 \geq 0. \end{array}$$

1. Write the dual of (P) .
2. Is (P) feasible? Is the dual feasible? What does this tell you?

Recap From Last Time & Today's Plan

Last time...

- **Separating Hyperplane Thm $\Rightarrow$ Farkas Lemma $\Rightarrow$ Strong duality**
- **Two motivating applications + implications of duality**

Agenda for today:

- Quick Recap
- Optimality conditions and primal/dual simplex
- Complementary slackness
- Global sensitivity & Shadow prices as marginal costs
- An important application: network revenue management / pricing

Polynomially-Sized CVaR Representation

- Suppose you are choosing weights for a portfolio $x \in X$:

$$\begin{aligned} & \text{maximize}_{x \in X} f(x) \\ & \text{s.t. } \text{CVaR}(R(x)) \geq b \end{aligned}$$

- The portfolio return $R(x)$ determined by x and n different scenarios for unknown returns
- CVaR is defined as the average over the k -smallest values of $R(x)$
- If payoffs in the n scenarios are $r^T(x) = [r_1(x), r_2(x), \dots, r_n(x)]$, the key constraint is:

$$\sum_{i=1}^k r_{[i]}(x) \geq k \cdot b, \tag{1}$$

where $r_{[1]}(x) \leq r_{[2]}(x) \leq \dots \leq r_{[n]}(x)$ is the sorted vector of payoffs under decision x .

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where $r_{[1]}(x) \leq r_{[2]}(x) \leq \dots \leq r_{[n]}(x)$ is the sorted vector of payoffs under decision x .

- **Claim.** Omit the dependency on x for simplicity; we have:

$$\sum_{i=1}^k r_{[i]} = \min_{y \in [0,1]^n} \left\{ r^T y : e^T y = k \right\}, \quad \text{where } e^T = [1, 1, \dots, 1]. \quad (2)$$

The intuition here is that it is optimal to set $y_i^* = 1$ exactly for the k smallest components $r_{[1]}, r_{[2]}, \dots, r_{[k]}$. This can be shown rigorously through a simple inter-change argument.

Polynomially-Sized CVaR Representation

- This claim implies that:

$$\sum_{i=1}^k r_{[i]}(x) \geq k \cdot b \quad \Leftrightarrow \quad \min_{y \in [0,1]^n} \left\{ \sum_{i=1}^n r^{\text{T}} y : e^{\text{T}} y = k \right\} \geq k \cdot b. \quad (3)$$

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- By strong duality,

$$\min_{y \in [0,1]^n} \left\{ \sum_{i=1}^n r^T y : e^T y = k \right\} = \max_{\lambda, t} \left\{ e^T \lambda + k \cdot t : \lambda + t \cdot e \leq r, \lambda \leq 0 \right\},$$

which implies that (3) is satisfied if and only if λ and t exist so that:

$$e^T \lambda + k \cdot t \geq b, \quad \lambda + t \cdot e \leq r, \quad \lambda \leq 0.$$

- Our original problem is therefore equivalent to the following problem in variables x, λ, t :

$$\begin{aligned} & \text{maximize}_{x, \lambda, t} && f(x) \\ & \text{s.t.} && x \in X \\ & && e^T \lambda + k \cdot t \geq b \\ & && \lambda + t \cdot e \leq r(x) \\ & && \lambda \leq 0. \end{aligned}$$

Application in Robust Optimization

- Consider an optimization problem with an uncertain constraint:

$$\begin{aligned} & \underset{x}{\text{minimize}} \quad f(x) \\ & \text{s.t.} \quad a^T x \leq b, \quad \forall a \in \mathcal{A} := \{a \in \mathbb{R}^n : Ca \leq d\} \end{aligned} \tag{4}$$

- $\mathcal{A}$ is polyhedral set; infinitely many constraints : “semi-infinite” LP.

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$$\min_{\lambda} \{ \lambda^T \mathbf{d} : \lambda^T \mathbf{C} = \mathbf{x}^T, \lambda \geq 0 \} \leq b$$

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- Our original problem is then equivalent to:

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- This is a polynomially-sized set of constraints in x, λ

Implications

Theorem (**Strong Duality**)

If $(\mathcal{P})$ has an optimal solution, so does $(\mathcal{D})$, and their optimal values are equal.

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		Dual		
		Finite Optimum	Unbounded	Infeasible
Primal	Finite Optimum	Possible	Impossible	Impossible
	Unbounded	Impossible	Impossible	Possible
	Infeasible	Impossible	Possible	?

Strong Duality and Theorems of Alternative

- Strong duality allows you to **prove** various “theorems of alternative”

Example (Farkas Lemma)

Prove that exactly one of the following is true:

- (i) $\exists x \geq 0$ such that $Ax = b$,
- (ii) $\exists \lambda$ such that $\lambda^T A \geq 0$ and $\lambda^T b < 0$.

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- Set up a (feasibility) problem that mirrors statement (i), and consider its dual.

$$\begin{array}{ll} (\mathcal{P}) \max & 0 \\ & Ax = b \\ & x \geq 0 \end{array} \qquad \begin{array}{ll} (\mathcal{D}) \min & \lambda^T b \\ & \lambda^T A \geq 0 \end{array}$$

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- (i) holds $\Rightarrow p^* = d^* = 0 \Rightarrow \lambda^T b \geq 0$ for any $\lambda : \lambda^T A \geq 0$, so (ii) cannot hold.
- (i) does **not** hold $\Rightarrow d^* = -\infty \Rightarrow \exists \lambda : \lambda^T b < 0$ and $\lambda^T A \geq 0$, so (ii) holds.

Optimality for Standard-Form LPs

$$(\mathcal{P}) \min c^T x$$

$$Ax = b, \quad x \geq 0$$

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$$\lambda^T A \leq c^T$$

- $(\mathcal{P})$ achieves optimality at a **basic feasible solution** x :

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 - If $B \subseteq \{1, \dots, n\}$ is a basis, the b.f.s. is: $x = [x_B, 0]$, $x_B = A_B^{-1}b$.
 - Simplex algorithm: feasibility and optimality for $(\mathcal{P})$ are given by:

$$\text{Feasibility-}(\mathcal{P}) : \quad x_B := A_B^{-1}b \geq 0 \quad (6a)$$

$$\text{Optimality-}(\mathcal{P}) : \quad c^T - c_B^T A_B^{-1} A \geq 0 \quad (6b)$$

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- $(\mathcal{D})$: same basis B can also be used to determine a **dual vector** λ :

$$\lambda^T A_i = c_i, \forall i \in B \Rightarrow \lambda^T = c_B^T A_B^{-1}, \forall i \in B.$$

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- λ is feasible in the dual if and only if:

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Primal optimality $\Leftrightarrow$ Dual feasibility

Simplex terminates when finding a dual-feasible solution!

Solve $(\mathcal{P})$ or $(\mathcal{D})$?

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Primal simplex

- maintain a **basic feasible solution**
- basis $B \subset \{1, \dots, n\}$
- stopping criterion: dual feasibility

Dual simplex

- maintain a dual feasible solution
- stopping criterion: primal feasibility
- different from primal simplex: works with an LP with inequalities

- How to choose $(\mathcal{P})$ or $(\mathcal{D})$?
- Suppose we have x^* , λ^* and must now solve a **larger** problem, i.e., with extra decisions or extra constraints.
- *Any preference between primal and dual simplex?*

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- *Any preference between primal and dual simplex?*
 - With extra decisions $x_e \Rightarrow$ **primal simplex** initialized with $[x^*, x_e = 0]$.
 - With extra constraints $A_e x = b_e \Rightarrow$ **dual simplex** initialized with $[\lambda^*, \lambda_e = 0]$.
- Modern solvers include primal and dual simplex and allow concurrent runs

Optimality Conditions and Complementary Slackness

Primal-Dual Pair of Problems

$$\begin{array}{llll} (\mathcal{P}) & \underset{x}{\text{minimize}} & c^T x & \\ & & Ax \leq b & \\ & & x \geq 0 & \\ \text{variables} & & x \in \mathbb{R}^n & \end{array}$$

$$\begin{array}{llll} (\mathcal{D}) & \underset{\lambda}{\text{maximize}} & \lambda^T b & \\ & & \lambda \geq 0 & \\ & & \lambda^T A \leq c^T & \\ \text{variables} & & \lambda \in \mathbb{R}^m. & \end{array}$$

Consider $x \in P, \lambda \in D$ (each feasible). How to check if they are **optimal**?

Optimality Conditions and Complementary Slackness

Primal-Dual Pair of Problems

$(\mathcal{P})$ minimize $c^T x$ $Ax \leq b$ $x \geq 0$ variables $x \in \mathbb{R}^n$	$(\mathcal{D})$ maximize $\lambda^T b$ $\lambda \geq 0$ $\lambda^T A \leq c^T$ variables $\lambda \in \mathbb{R}^m$
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Theorem (Complementary Slackness)

$x \in P$ and $\lambda \in D$ are **optimal** solutions for $(\mathcal{P})$ and $(\mathcal{D})$, respectively, **if and only if**:

$$\lambda_i (a_i^T x - b_i) = 0, i = 1, \dots, m$$

$$(\lambda^T A_j - c_j) x_j = 0, j = 1, \dots, n.$$

- Follows from primal/dual feasibility and $c^T x = b^T \lambda$

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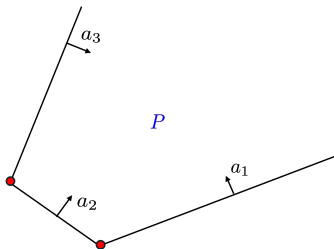
- Follows from primal/dual feasibility and $c^T x = b^T \lambda$
- Interesting insight: **non-binding constraint** $\Rightarrow$ dual variable is **zero**

Representation of Polyhedra

Important consequence of duality: alternative representation of all polyhedra

Definition

Consider a nonempty polyhedron $P = \{x \in \mathbb{R}^n : Ax \geq b\}$. Then:



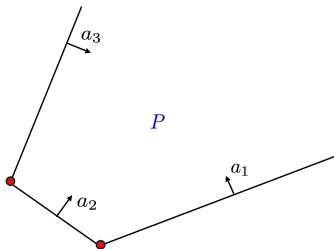
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1. $\mathcal{C} := \{d \in \mathbb{R}^n : Ad \geq 0\}$ is called the **recession cone** of P .
2. Any $d \in \mathcal{C}$ with $d \neq 0$ is called a **ray** of P .



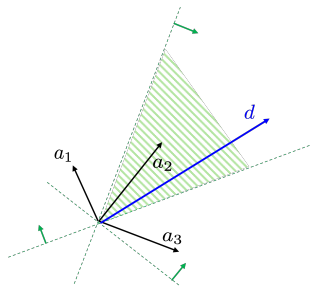
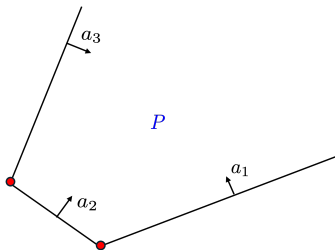
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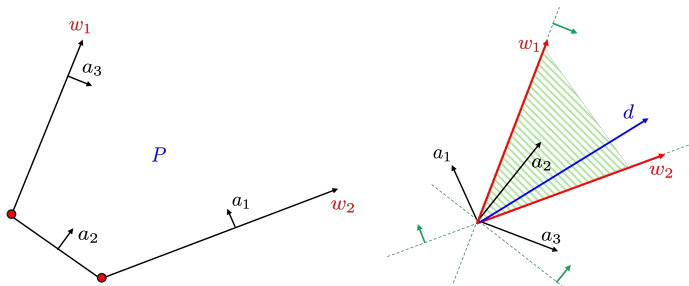
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3. Any ray d that satisfies $a_i^\top d = 0$ for $n - 1$ linearly independent a_i is called an **extreme ray** of P .



Representation of Polyhedra

Theorem (Resolution Theorem)

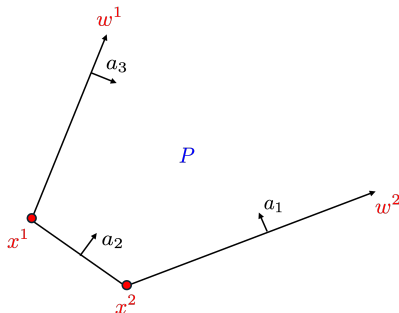
Let $P = \{x \in \mathbb{R}^n : Ax \geq b\}$ be a non-empty polyhedron, $x^1, x^2, \dots, x^k$ be its **extreme points**, and $w^1, w^2, \dots, w^r$ be its **extreme rays**. Then,

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$$P = \text{conv}(\{x^1, \dots, x^k\}) + \text{cone}(\{w^1, \dots, w^r\})$$
$$= \left\{ \sum_{i=1}^k \mu_i x^i + \sum_{j=1}^r \theta_j w^j : \mu \geq 0, e^T \mu = 1, \theta \geq 0 \right\}.$$

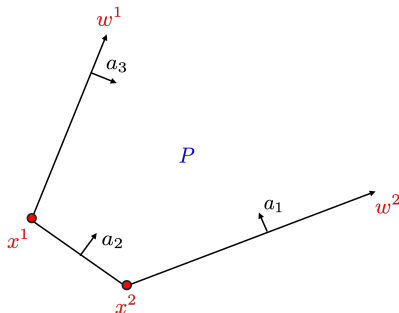


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Note: It is **not** “easy” (i.e., poly-time) to switch between these representations

Dual Variables **As Marginal Costs**

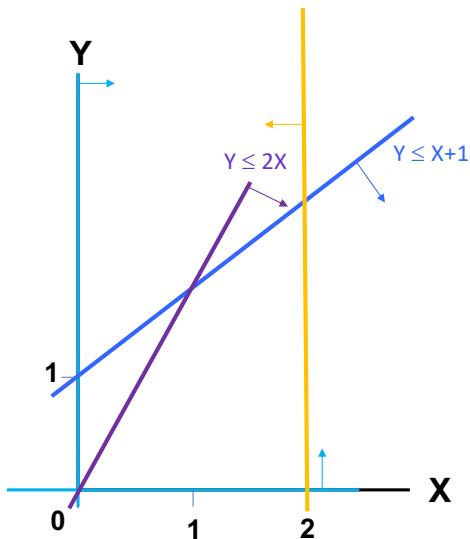
$$\begin{array}{ll} (\mathcal{P}) \min c^T x & (\mathcal{D}) \max \lambda^T b \\ Ax = b, \ x \geq 0 & \lambda^T A \leq c^T \end{array}$$

- Solved the LP and obtained x^* and λ^*
- Want to show that λ^* is the **gradient of the optimal cost with respect to b** “almost everywhere”
- Related to **sensitivity analysis**
How do the optimal value and solution depend on problem data A, b, c ?

Sensitivity: A Simple Example

Maximize Y

Subject to: $Y \leq 2X$
 $Y \leq X+1$
 $X \geq 0, Y \geq 0$
 $X \leq 2$

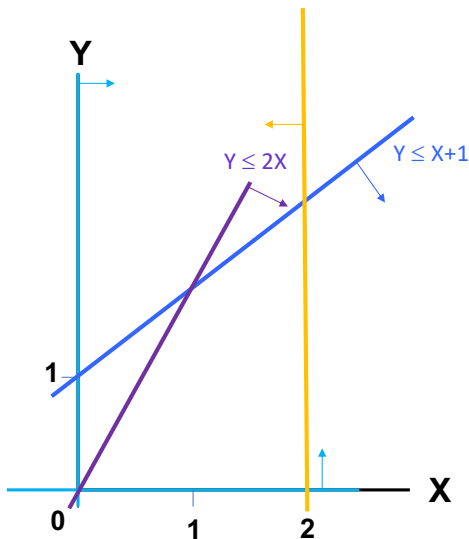


Sensitivity: A Simple Example

Maximize Y

Subject to: $Y \leq 2X$
 $Y \leq X+1$
 $X \geq 0, Y \geq 0$
 $X \leq 2$
 $X \leq a$

For the last constraint $X \leq a$,
what is the *shadow price*
i.e., rate of change in the
optimal value when we change
the constraint r.h.s. a ?

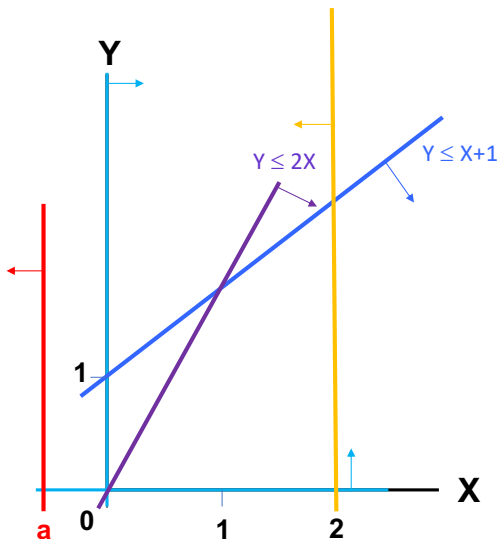


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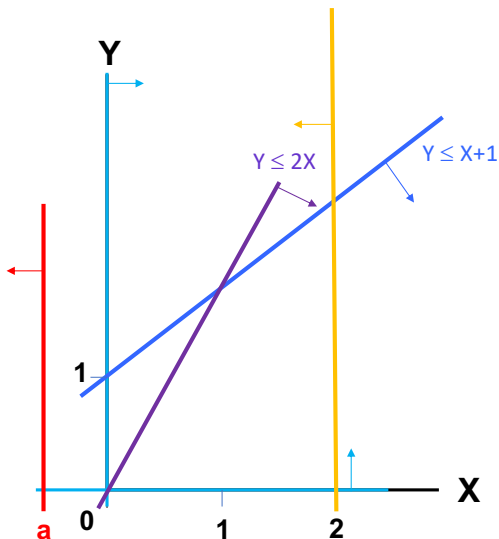
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- Infeasible!

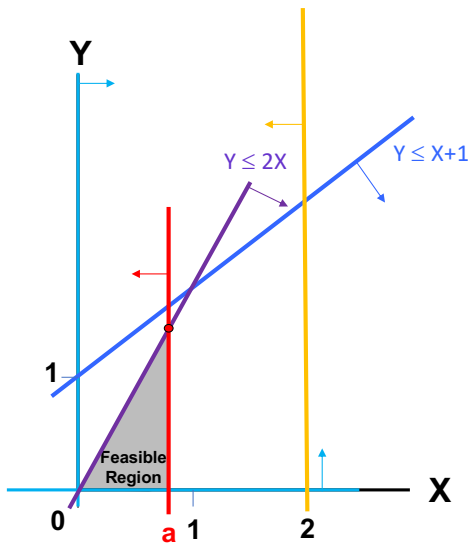


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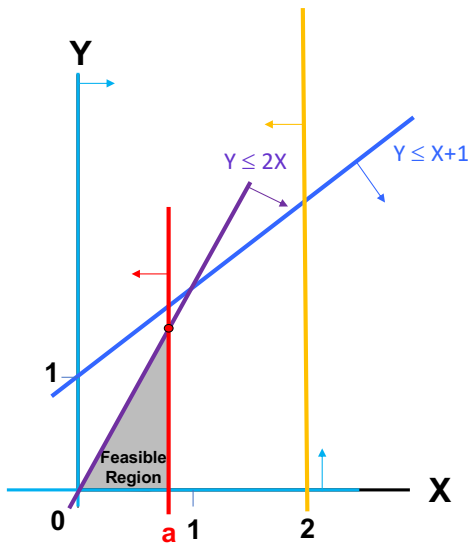
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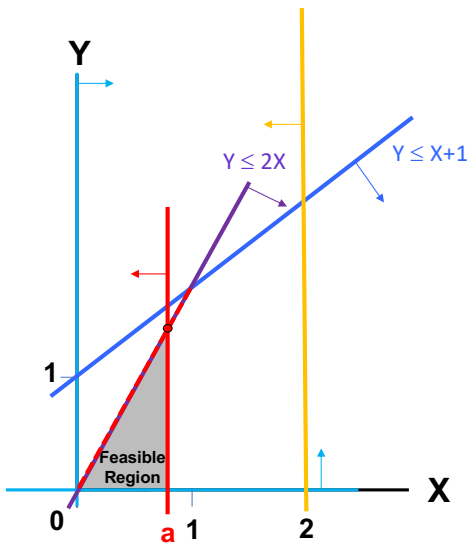
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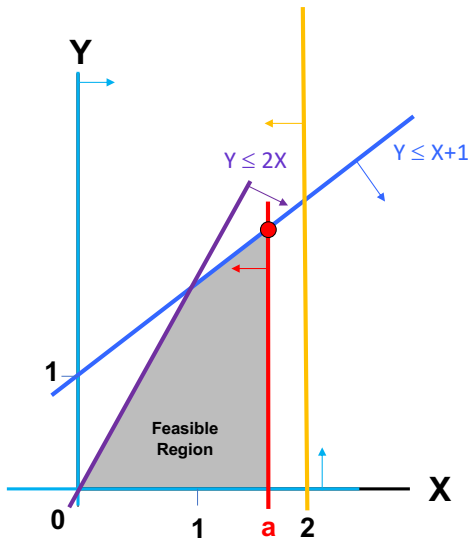


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Sensitivity: A Simple Example

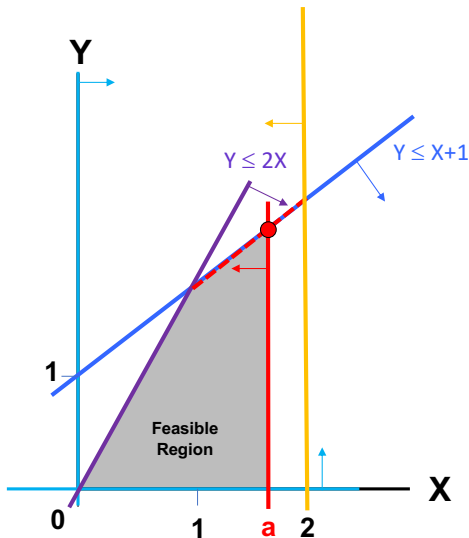
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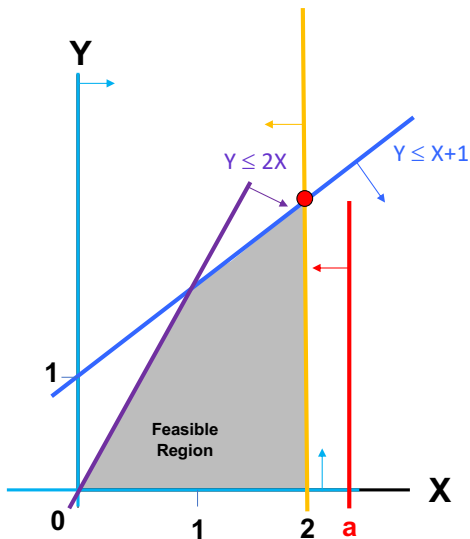


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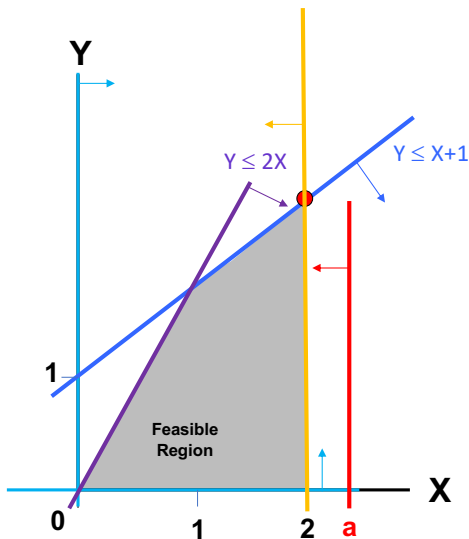
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Maximize Y

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If $a > 2$:

- Shadow price = 0

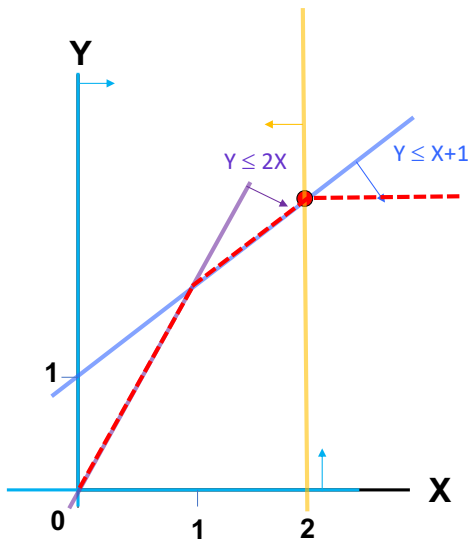


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Note how the objective depends on a overall



Global Dependency On b, c

$$\begin{array}{ll} (\mathcal{P}) \min c^T x & (\mathcal{D}) \max \lambda^T b \\ Ax = b, \ x \geq 0 & \lambda^T A \leq c^T \end{array}$$

- What to show that the **optimal value** (when finite) **as a function of b** is
- What to show that the **optimal value** (when finite) **as a function of c** is

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- What to show that the **optimal value** (when finite) **as a function of b** is piecewise linear and **convex**
- What to show that the **optimal value** (when finite) **as a function of c** is piecewise linear and **concave**

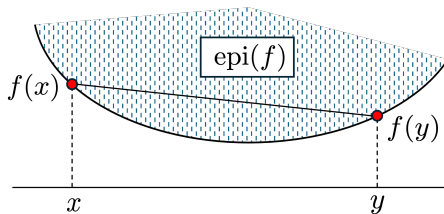
Convex and Concave Functions

Definition

$f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is **convex** if X is a convex set and

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y), \quad \forall x, y \in X \text{ and } \lambda \in [0, 1]. \quad (8)$$

A function is **concave** if $-f$ is convex.



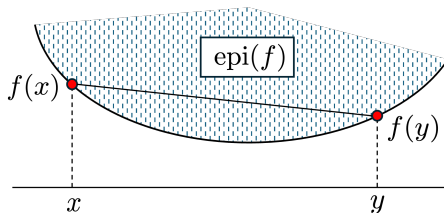
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Equivalent definition in terms of **epigraph**:

$$\text{epi}(f) = \{(x, t) \in X \times \mathbb{R} : t \geq f(x)\} \quad (9)$$

f is convex if and only if $\text{epi}(f)$ is a convex set.

Global Dependency On b

- Let $P(b) := \{x \in \mathbb{R}^n : Ax = b, x \geq 0\}$ denote the feasible set of the primal
- Let $S := \{b \in \mathbb{R}^m : P(b) \neq \emptyset\}$: right-hand-side values that yield a feasible primal
- Let $p^*(b)$ denote the optimal objective; assume $p^*(b) > -\infty$ (i.e., dual is feasible)

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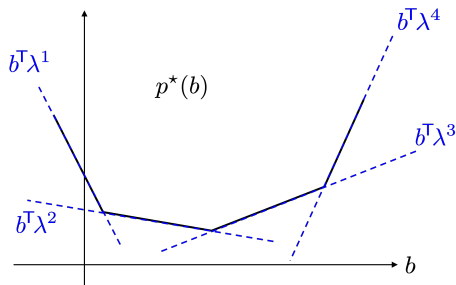
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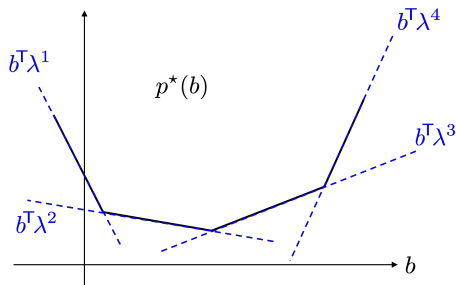


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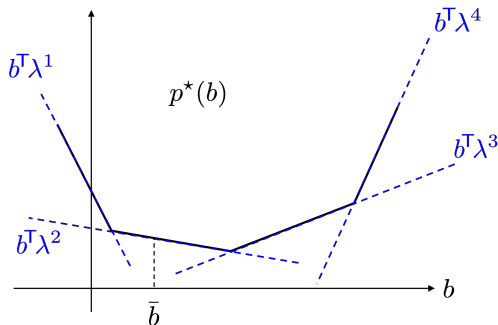
How to prove $p^(b)$ convex?*

$$\text{epi}(p^*) = \cap_{i=1, \dots, r} \text{epi}(b^T \lambda^i)$$

is the intersection of convex sets, so it is convex.

Global Dependency On b - Implications

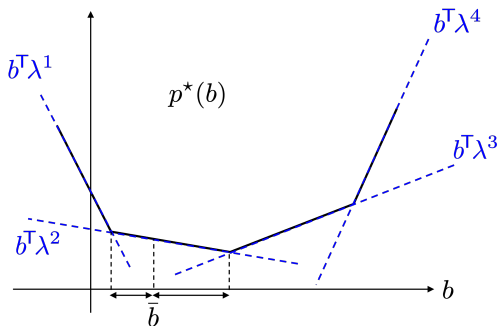
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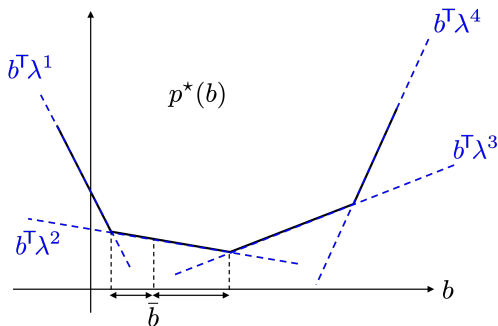
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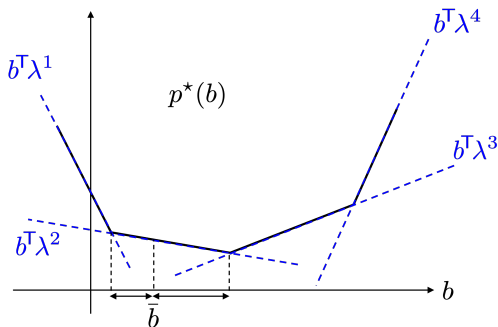


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- λ_i^* acts as a **marginal cost** or **shadow price** for the i -th constraint r.h.s. b_i
- λ_i allows estimating **exact change in p^* in a range around $\bar{b}_i$**
- Modern solvers give **direct access to λ_i^* and the range**

Gurobipy: for constraint c , the attribute $c.Pi$ is λ_i^* and the range is from $c.SARHSLow$ to $c.SARHSUP$

Global Dependency On b - Implications

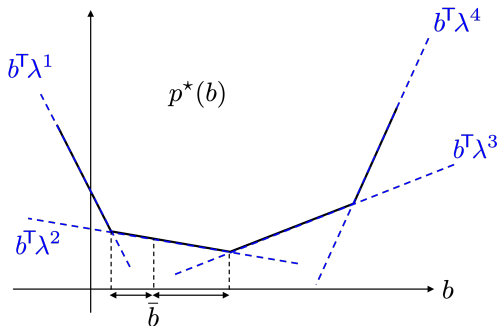
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- All such λ^i are valid **subgradients** of p^*

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Definition (Subgradient.)

$f : S \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ convex function. A vector $g \in \mathbb{R}^n$ is a **subgradient** of f at $\bar{x} \in S$ if

$$f(x) \geq f(\bar{x}) + g^T(x - \bar{x}), \quad \forall x \in S.$$

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- If for some c the LP has a **unique** optimal solution x^* , then d^* is linear in the vicinity of c and its gradient is x^* .
- The optimal primal solution x^* **is a shadow price for the dual constraints**
- x^* remains optimal for a range of change in each objective coefficient c_j
- Modern solvers also allow obtaining the range directly
Gurobi: attributes **SAObjLow** and **SAObjUp** for each decision variable

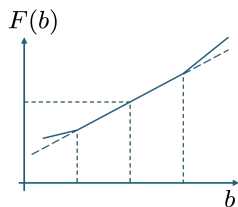
Signs of Dual Variables Revisited

- There is a direct connection between:
 - the **optimization problem** (max/min)
 - the **constraint type** ($\leq$, $\geq$)
 - the **signs of the shadow prices**
- Given two of these, can figure out the third one!
- *What is the sign of the shadow price for a ...*
 - $\leq$ constraint in a **minimization** problem ?
 - $\geq$ constraint in a **minimization** problem ?
 - $\leq$ constraint in a **maximization** problem ?
 - $\geq$ constraint in a **maximization** problem ?
- *What is the dependency of the optimal objective on the r.h.s. of a ...*
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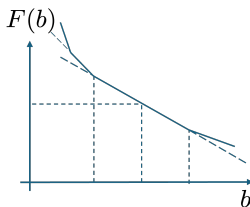
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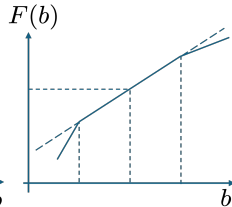
$\min, \geq b$
dual ≥ 0
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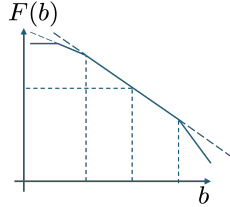
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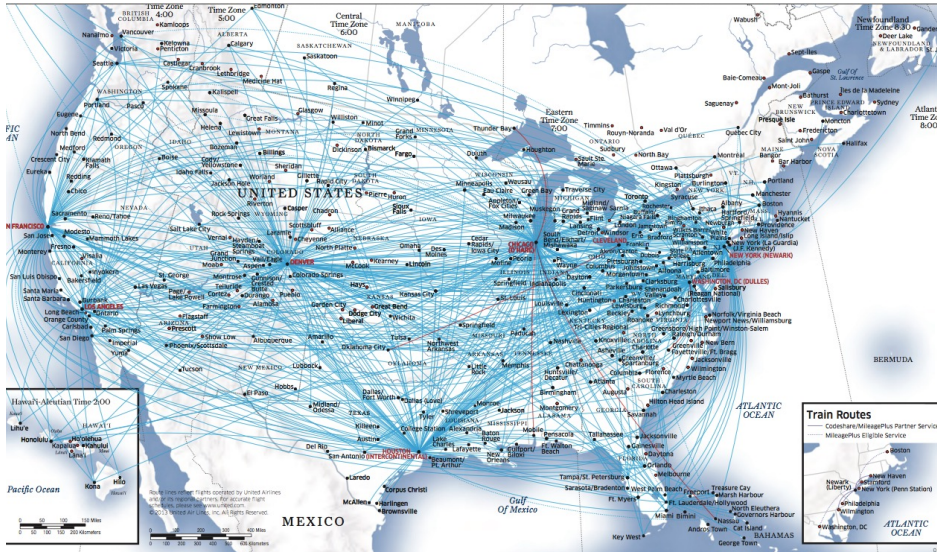
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Real-World Hub and Spoke Airline Network



Source: www.united.com

Airline Revenue Management (RM)

- **Strategic RM**

- Determine several price points for various itineraries
- “Product” or “itinerary”: origin, destination, day, time, various restrictions, ...
 - E.g., JFK – ORD – SFO, 10:30am on Oct 12, 2024, Economy class Y fare
- Typically done by (or in conjunction with) marketing department
 - Market segmentation; competition

- **Tactical RM (“yield management”)** decides **booking limits**

- A *booking limit* determines how many seats to reserve for each “product”
- RM not based on setting prices, but rather changing availability of fare classes
- Legacy due to original IT systems used (e.g., SABRE)

Airline RM

Hub: Chicago ORD

Two planes  

Westbound flights for
some day in the future

SFO



ORD



LAX



BOS



JFK



Airline RM

Flight segments (legs)

SFO



ORD



LAX



BOS



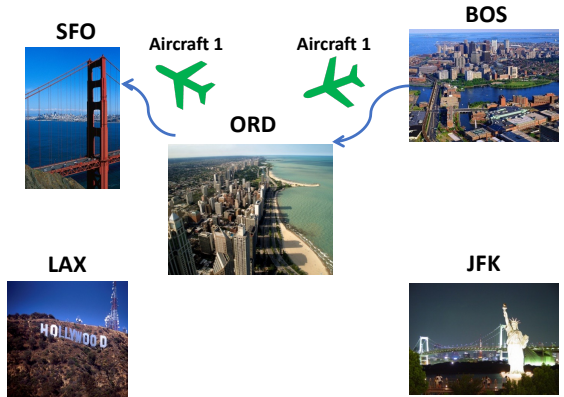
JFK



Airline RM

Flight segments (legs)

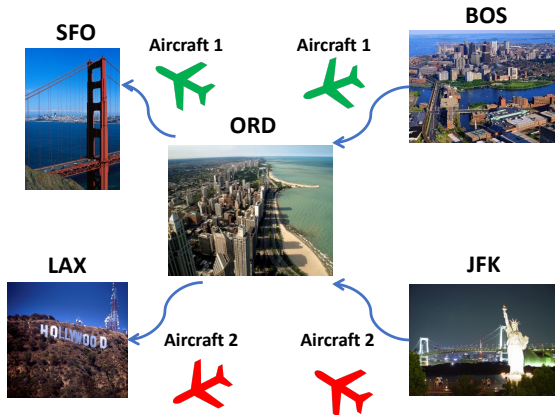
- Aircraft 1: 
 - BOS-ORD in the morning
 - ORD-SFO in the afternoon



Airline RM

Flight segments (legs)

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- **Aircraft 2:** 
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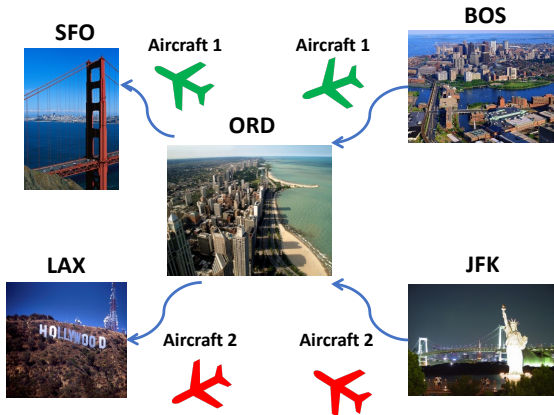
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Itineraries

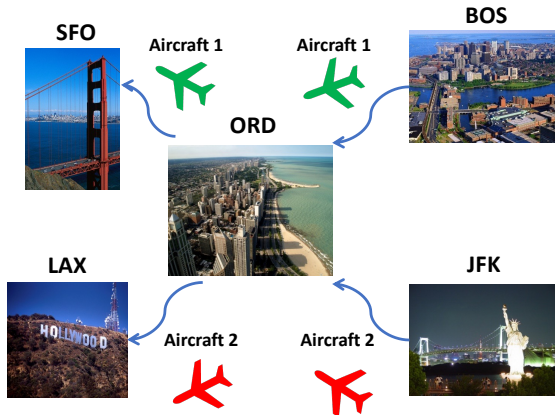
Origin-Destination	Q_Fare	Y_Fare
BOS_ORD	\$200	\$220
BOS_SFO	\$320	\$420
BOS_LAX	\$400	\$490
JFK_ORD	\$250	\$290
JFK_SFO	\$410	\$540
JFK_LAX	\$450	\$550
ORD_SFO	\$210	\$230
ORD_LAX	\$260	\$300



Airline RM

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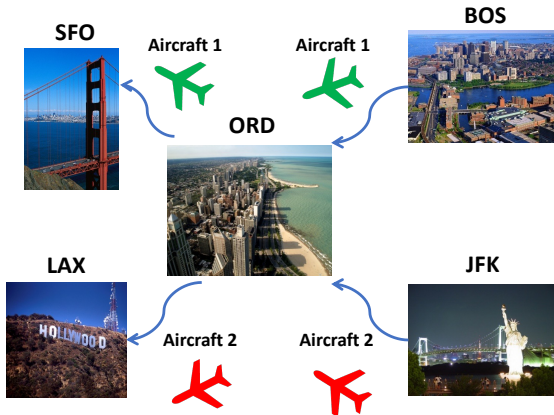
Itineraries

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BOS_ORD	\$200	\$220	25	20
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Airline RM

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Resources needed

	BOS_ORD	BOS_SFO	BOS_LAX	JFK_ORD	JFK_SFO	JFK_LAX	ORD_SFO	ORD_LAX
Flight leg								
BOS_ORD_Leg	1	1	1	0	0	0	0	0
JFK_ORD_Leg	0	0	0	1	1	1	0	0
ORD_SFO_Leg	0	1	0	0	1	0	1	0
ORD_LAX_Leg	0	0	1	0	0	1	0	1

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- Airline operates a set F of direct flights in its (hub-and-spoke) network

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Resource matrix A :	Flight leg 1	1	0	...	1
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- Goal: decide how many itineraries of each type to sell to maximize revenue

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